Forecasting for the Grain the Decision Is Made At: A Retail World Model on M5, Dominick's and Breakfast at the Frat

Karthik Iyer

Correspondence: karthik.iyer488@gmail.com ORCID: 0009-0004-0593-1602

Abstract. Which forecasting model is best depends on the level of the product and store hierarchy at which the forecast is judged, and on how much of the sales the plan (price, promotion, events) moves. We compare a world model, which carries each item's state forward one period at a time under the plan, with a direct lift model, LightGBM, a graph model, seasonal naive, and the pretrained TimesFM 2.5 with and without the plan, on three public data sets: M5 (four 28-day windows), six Dominick's categories and Breakfast at the Frat. Every model is scored at every level of the hierarchy, and every model and change that lost is reported. On M5, TimesFM given only each item's sales history is the most accurate model item by store in all four months (0.836 to 0.885), but forecasts 5% to 10% below total sales; the world model is the most accurate at the chain total in two of the four months and LightGBM in one. The official M5 test month is the only one of the four in which LightGBM is the most accurate over all levels, so a single test month would have given the wrong ranking. On the weekly, promotion-driven data, TimesFM without the plan forecasts 29% to 63% of promoted-week sales and is behind both plan-aware trained models item by store in six of seven data sets; given the plan, it forecasts 60% to 113% and becomes the most accurate model in one category. For an item whose first price cut in a year sold 120 units, the world model forecast 121, 600, 243 across three seeds, against about 3 for LightGBM and 8 for TimesFM: plan-aware models respond in the right direction, and the data do not fix the size. A retailer should choose its forecasting model by the decision the forecast feeds, scored at that decision's level and over several periods.

Keywords: retail forecasting; hierarchical evaluation; promotions; world models; foundation models; M5.

1 Introduction

A retailer uses one sales forecast for several decisions, and they are made at different levels. Price changes, promotions, supplier funding and store orders are decided for one item in one store, week by week. Sales, unit and margin plans are set and reviewed for a category or a whole chain, by month. A forecast can be accurate item by store and still miss the month's category total by several percent, and the reverse.

Forecasting competitions and most published comparisons rank models on one combined score over one test period. The M5 competition [1] already weights its score across twelve levels of the Walmart hierarchy, but reports one number per entry, for one 28-day month. This paper asks three questions that a single number hides: which model is most accurate at each level, whether that answer holds from one month to the next, and how much it depends on whether the model sees the plan.

We answer them with seven models on three public data sets and eleven final windows, scoring every model at every level of each data set's hierarchy. The models range from seasonal naive to a pretrained time-series foundation model (TimesFM 2.5 [15]) used without training on our data, and include a world model of our own: each item's recent history is turned into a state, and that state is carried forward one period at a time under the plan, so that what is planned for one week can change the forecast for the weeks after it (Figure 1).

The paper finds and contributes the following.

1. **The most accurate model depends on the level.** On M5, TimesFM given only sales history is the most accurate item by store in all four months, by 0.005 to 0.014 against the world model and 0.018 to 0.037 against LightGBM, while models that see the plan are more accurate at every level from store by department up to the chain total (§5.1).

2. **One test month is not enough.** The official M5 test month is the only one of four in which LightGBM is the most accurate over all levels. The world model is lowest in March and, by 0.002, in April, and the graph model on the practice window (§5.2).

3. **The value of the plan depends on how much of the sales it moves.** Where prices rarely change (M5), giving TimesFM the plan changes its item-by-store error by at most 0.002. Where promotions carry 18% to 61% of a category's units (Dominick's), TimesFM

without the plan forecasts 29% to 63% of promoted-week sales (§5.3).

4. **Plan-aware models respond to price changes that history has not shown, with a size the data do not fix.** For an item's first price cut in a year the world model forecast 121, 600, 243 units across seeds against 120 sold, where models working from history forecast under 10 (§5.4).
5. **Every model and change that lost is reported**, with its cause where one was measured (§5.5, Table 5).

2 Related work

Retail forecasting and promotions. Fildes, Ma and Kolassa [2] review retail forecasting and note that most studies evaluate at one level of aggregation, usually the item. Promotion effects have long been modelled from store scanner data: PromoCast forecast promoted-week sales from the planned event [3], van Heerde, Leeflang and Wittink measured the dips before and after a promotion [4], and Ma, Fildes and Huang showed that promotional information from within and across categories improves item forecasts [5]. Dominick's, used here, is the data set behind much of this work [6].

Hierarchies and evaluation. Forecasts that must add up across a hierarchy are studied as hierarchical forecasting [8, 9]. The M5 competition scored entries with a weighted, scaled error averaged over twelve levels [1], the measure we use. Its leading entries were ensembles of gradient-boosted trees [10, 11]; our LightGBM follows their design without the ensembles. Tashman [12] sets out why a comparison should use several forecast origins rather than one test period, which §5.2 confirms on M5.

Neural and pretrained forecasters. Global neural models trained across many series, such as DeepAR [13] and the Temporal Fusion Transformer [14], take known future inputs such as price. Pretrained foundation models, among them TimesFM [15] and Chronos [17], forecast series they were not trained on from their history alone. TimesFM 2.5 also offers a covariate method that fits a linear model on future inputs and forecasts the remainder with the pretrained model [16], which we use to give it the plan. Graph neural networks share information between related series; our graph model uses GraphSAGE-style message passing [18].

World models. A world model learns a state of its environment and how that state moves under actions, so that the effect of a planned action can be rolled forward before it is taken [19, 20, 21]. In retail the actions are the plan: prices, promotions, displays. Our world model applies that structure to sales forecasting with a recurrent transition [22] on top of a transformer encoder [23]. We compare it with the same model without the roll-forward, so that the effect of carrying the state is measured separately from the rest of the design.

Table 1: Seven models: five see the plan, two forecast from history alone. Trained: fitted on the data set; the TimesFM variants are used as published.

Model	Plan	Trained	What it is
World model	yes	yes	Item state rolled forward under the plan
Lift model	yes	yes	Same encoder; baseline times plan lift; no roll-forward
LightGBM	yes	yes	One tree model per store, Tweedie loss
Graph model	yes	yes	LightGBM's inputs; message passing between related series
TimesFM with the plan	yes	no	TimesFM 2.5 with its covariate method
TimesFM	no	no	TimesFM 2.5, 512 periods of units
Seasonal naive	no	no	Sales one season earlier

3 Models

Seven models are compared (Table 1). Five see the plan for the forecast periods and two do not.

World model. For item i in one store, an encoder (a transformer over the last L periods: 112 days on M5, 26 weeks on Dominick's and Frat) reads sales and price, each scaled by the item's recent level ℓ_i, together with the calendar, and gives a state s_t. A gated recurrent transition f carries the state forward under the plan p_{t+h} for each future period,

$$s_{t+h}(p) = \mathrm{LN}(f(p_{t+h},\, s_{t+h-1}(p))), \quad s_t(p) = s_t, \qquad (1)$$

where LN is layer normalisation. The plan holds the period's price as a share of the item's regular price (its highest price in the last 12 periods) and the recorded levers: event and SNAP days on M5; promotion on Dominick's; promotion, display, feature and price-reduction-only on Frat. Expected sales are a baseline under the "nothing planned" plan p^0 (regular price, no lever) times a lift,

$$\begin{aligned}\log \mu_{i,t+h} = {} & \log \ell_i + b(s_{t+h}(p^0), p^0_{t+h}) \\ & + g(s_{t+h}(p), p_{t+h}) - g(s_{t+h}(p^0), p^0_{t+h}),\end{aligned} \qquad (2)$$

with b and g small networks that also take the item's labels and the horizon step, and the log forecast bounded

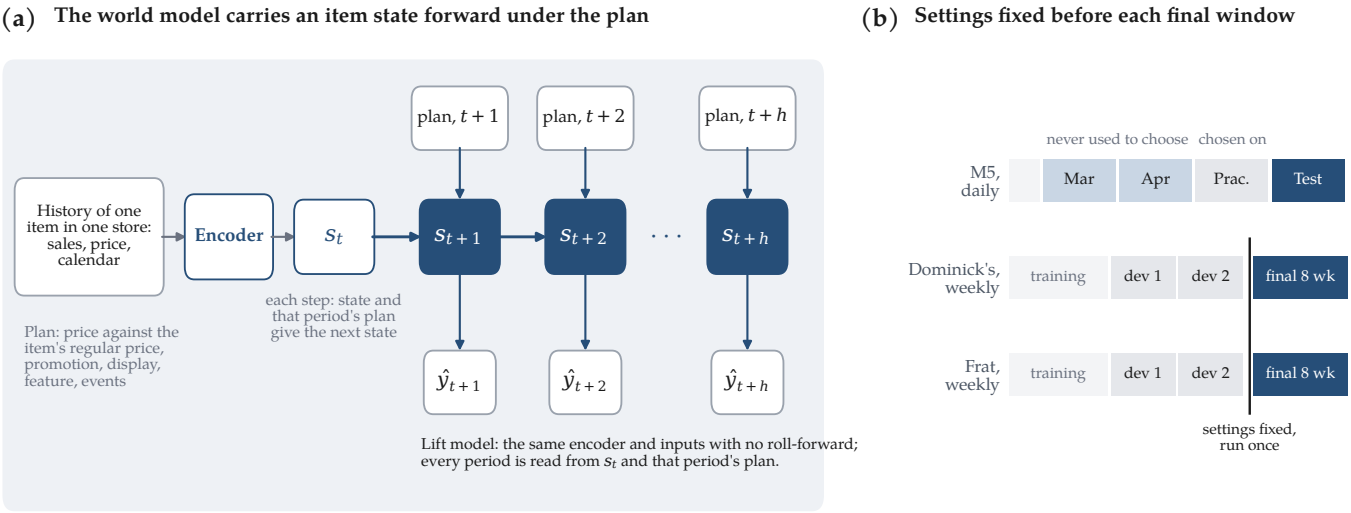


Figure 1: The world model carries each item's state forward under the plan, and every final window was run once with settings fixed beforehand. (**a**) An encoder turns the recent history of one item in one store into a state s_t. A transition step takes the state and one period's plan and gives the state one period later; each period's sales are read from the state as it stands after the plan so far. The lift model shares the encoder and inputs and reads every period from s_t. (**b**) On M5, settings were chosen on the practice window (25 April to 22 May 2016); the two earlier months were not used for any choice. On Dominick's and Frat, settings were chosen on two development windows before the final 8 weeks.

to $[-10, 10]$. The lift is exactly zero when nothing is planned. Training minimises a Tweedie loss on sales relative to ℓ_i. On Dominick's, store-week groups of items are rescaled so the group total rises by a learned share of the summed lift (the rest is switching between items), and three fits are averaged. Model sizes, training steps and every other setting are in the run records (§9).

Lift model. The same model without the roll-forward: $s_{t+h}(p) = s_t$ for every h. The two share inputs and training, so the difference between them is the roll-forward.

Comparison models. *LightGBM* [10]: one direct model per store, Tweedie loss, lagged and rolling sales, price against its recent highest, calendar and plan inputs. *Graph model*: LightGBM's inputs, with one round of GraphSAGE-style message passing [18] between the same item in other stores and items of the same department in the same store, Tweedie loss. *TimesFM* [15, 16]: the 200M-parameter TimesFM 2.5 checkpoint given each series' last 512 periods of units; we use its mean output, because its default median forecasts 27% to 31% below M5 totals (Table 5). *TimesFM with the plan*: the same model through its covariate method, a linear model on the plan inputs fitted over groups of 2,048 series, then TimesFM on what that model leaves. *Seasonal naive*: sales one season earlier (a week on M5).

What the state holds. A probe on the frozen world model shows that the carried state takes in the plan (Figure 2). The day's discount can be read back from the rolled state with R^2 0.58 to 0.83 across three M5 windows, against 0.00 to 0.02 from the state before the window, and with 0.46 to 0.59 seven days later. This shows that the plan is in the state; it does not show that the forecast uses it well, which the results test.

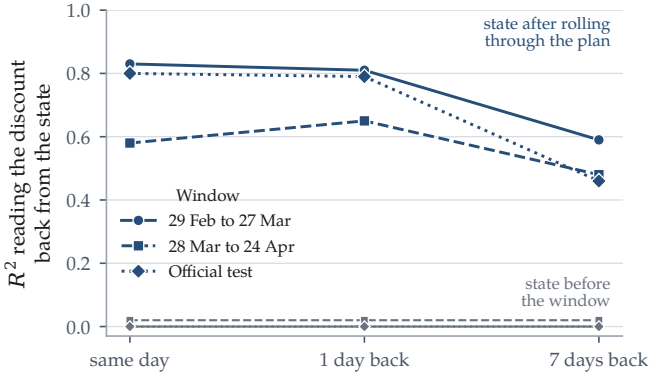


Figure 2: The rolled state holds the plan, and still holds most of it a week later. R^2 of a ridge read-out of the day's discount (price against regular price) from the frozen world model's state, fitted on half of 6,000 item-store pairs and scored on the other half. Blue: the state after rolling through the window's plan. Grey: the state before the window, which has not seen the plan. Seed 1 in three M5 windows.

4 Experimental setup

Data. All three data sets are public (Table 2). M5 holds daily Walmart sales of 3,049 items in 10 stores [1]. Dominick's Finer Foods holds weekly scanner data by store; we use six categories chosen before any final run [6]. Breakfast at the Frat holds weekly sales of 55 breakfast

Table 2: Three public data sets, all scored item by store and at every level above it.

Data	Series	Plan inputs	Final windows
M5 (daily)	30,490	Price, event days, SNAP days	Four 28-day windows, 29 Feb to 19 Jun 2016
Dominick's (weekly)	Six categories	Price, promotion	Last 8 weeks of each
Frat (weekly)	3,909	Price, promotion, display, feature, price reduction only	Last 8 weeks, 16 Nov 2011 to 4 Jan 2012

items in 77 stores with the promotion, display and feature for each week [7]. No raw data is redistributed.

Measure. We use the weighted root mean squared scaled error (WRMSSE) of M5 [1] at each level of the hierarchy: 12 levels on M5 (from the chain total to item by store) and six on Dominick's and Frat (total, category, store, category by store, item, item by store). Each series' error is scaled by its own in-sample one-step error, and series are weighted by their dollar sales over the last weeks of training. We report the score over all levels, the chain total and item by store, and the ratio of total forecast to total sales. Lower error is better.

Protocol. Settings of the world model, the lift model and LightGBM were chosen on the M5 practice window (25 April to 22 May 2016) and on two development windows before the final 8 weeks of Dominick's and Frat, then fixed and run once on the final windows (Figure 1b). Two earlier M5 months, 29 February to 27 March and 28 March to 24 April 2016, were not used for any choice; each is forecast from data up to the day before it starts. Trained models were run with three seeds and we report the mean and the range. TimesFM, TimesFM with the plan and the graph model were added after the final tests as comparisons, with settings fixed before their first run and no development windows of their own.

5 Results

The results follow the order of the contributions: the level at which a model is judged (§5.1), the number of months it is judged on (§5.2), the plan (§5.3, §5.4), and the failures (§5.5). Table 3 and Table 4 hold every final score.

5.1 The most accurate model depends on the level

On M5, TimesFM given only each item's sales history is the most accurate model item by store in all four months (Figure 3a): 0.836, 0.828, 0.828 and 0.885, against 0.851, 0.833, 0.834 and 0.897 for the world model and 0.873, 0.846, 0.847 and 0.905 for LightGBM. Among the trained models, the world model and the lift model are ahead of LightGBM and the graph model item by store in every month, and in both confirmation months every world-model seed is ahead of every LightGBM seed.

At the total the order reverses (Figure 3b). TimesFM forecasts 5% to 10% below total sales in every month, so its total error is higher than seasonal naive's in three of the four months (0.830 against 0.658 in March) and level with it on the test month (0.558 against 0.560). The world model has the lowest total error in March (0.398) and April (0.283), the graph model on the practice window (0.455) and LightGBM on the test month (0.247).

Figure 4 shows where the order changes. From the chain total down to store by department, the plan-aware models are ahead of TimesFM in every month. At the three item levels (item, item by state, item by store) TimesFM is the most accurate model in three of the four months, and item by store in all four. A model that is best for store orders on M5 is therefore not the model to use for a category plan. The combined M5 score weights the twelve levels equally, and nine of them are above the item, so a ranking on it mostly reflects the upper levels.

5.2 One test month misleads

Over all twelve levels, the official M5 test month (23 May to 19 June 2016) is the only one of the four in which LightGBM is the most accurate model (Figure 3c): 0.547 against 0.638 for the world model, level with the 10th-ranked entry of the competition (0.547). In March the world model is ahead, 0.642 (seeds 0.632 to 0.648) against 0.709 (0.706 to 0.713); in April the two are level, 0.555 against 0.557; and on the practice window the graph model is lowest (0.641), with the world model at 0.656 and LightGBM at 0.715.

Between the world model and LightGBM, the ranking follows the level of the total forecast (Figure 5a). LightGBM forecasts 3% to 6% below total sales in March and April and 2% below on the test month; the world model is within 1% in March and April and 2.6% above on the test month. In each of these three months, the one of the two whose total is closer to sales has the lower total error and the lower error over all levels. Grouping April's item-store pairs by their March sales rate (Figure 5b), both models are within 6% of sales in every group that sold in March; LightGBM is under in every group and the world model is closer to sales on 50.6% to 54.2% of pairs in each. The world model's weak spot is pairs that sold nothing in March, where it forecasts 62% above April sales.

Table 3: On M5 the best model depends on the level scored and on the month. WRMSSE on the four 28-day windows (lower is better), mean over seeds; seeds per window in the second column. All: the 12 levels together; Total: the chain total; I×S: item by store. The lowest value in each column is in bold. TimesFM with the plan has no recorded total. The lift model's April mean includes one failed seed (2.603 overall; §5.5).

		29 Feb to 27 Mar			28 Mar to 24 Apr			25 Apr to 22 May (practice)			23 May to 19 Jun (test)		
Model	**Seeds**	All	Total	I×S	All	Total	I×S	All	Total	I×S	All	Total	I×S
World model	3/3/3/3	**0.642**	**0.398**	0.851	**0.555**	**0.283**	0.833	0.656	0.535	0.834	0.638	0.379	0.897
Lift model	3/3/3/3	0.664	0.427	0.846	1.246	0.964	0.834	0.713	0.623	0.832	0.663	0.368	0.896
LightGBM	3/3/1/3	0.709	0.578	0.873	0.557	0.351	0.846	0.715	0.659	0.847	**0.547**	**0.247**	0.905
Graph model	3/3/3/3	0.778	0.621	0.878	0.644	0.493	0.845	**0.641**	**0.455**	0.851	0.673	0.412	0.909
TimesFM	1/1/1/1	0.854	0.830	**0.836**	0.819	0.786	**0.828**	0.896	0.930	**0.828**	0.729	0.558	0.885
TimesFM with the plan	1/1/1/1	0.858	–	0.837	0.772	–	0.829	0.929	–	0.830	0.719	–	**0.883**
Seasonal naive	1/1/1/1	0.889	0.658	1.121	0.923	0.668	1.126	0.870	0.751	1.100	0.847	0.560	1.176

Table 4: On the promotion-driven weekly data, models that see the plan lead, and which one leads changes by category. WRMSSE on the final 8 weeks (lower is better). World model, lift model and LightGBM: mean of three seeds, settings fixed on the development windows. TimesFM and TimesFM with the plan: one run each, added after the final tests. The lift model was run on Frat only. The lowest value in each row and group is in bold.

	All levels					Item by store				
Data	World	Lift	LightGBM	TimesFM	TimesFM+plan	World	Lift	LightGBM	TimesFM	TimesFM+plan
Frat	0.681	0.714	**0.549**	0.965	0.686	0.665	0.677	**0.608**	0.827	0.641
Dominick's										
Cheese	0.897	–	0.708	0.829	**0.545**	0.715	–	0.702	0.792	**0.699**
Snack crackers	0.448	–	**0.412**	0.468	0.475	0.608	–	**0.596**	0.640	0.614
Soft drinks	**0.830**	–	0.989	1.070	0.974	0.734	–	**0.652**	0.895	0.780
Bottled juice	0.987	–	0.994	**0.940**	0.990	0.816	–	**0.750**	0.782	0.754
Analgesics	**0.867**	–	0.978	1.009	0.889	**0.713**	–	0.714	0.790	0.742
Frozen entrees	0.485	–	**0.478**	0.563	0.566	0.557	–	**0.529**	0.682	0.599

A comparison on the official test month alone would have ranked LightGBM first. On three of the four months it is not.

5.3 How much the plan matters

The plan matters in proportion to how much of the sales it moves. On M5, where prices change rarely, giving TimesFM the plan changes its item-by-store error by at most 0.002 in any month (0.837 against 0.836 in March). On Dominick's and Frat, where promotions drive the weeks, it changes a great deal (Figure 6).

Without the plan, TimesFM is behind both the world model and LightGBM item by store on Frat and in five of six Dominick's categories; in bottled juice it is between them (Figure 6b). This is the reverse of M5. The reason shows in the promoted weeks (Figure 7a): TimesFM forecasts 29% to 63% of what sold in a promoted week, because nothing in its input says the promotion is coming.

Given the plan, TimesFM forecasts 60% to 113% of promoted-week sales, and its overall error improves on Frat (0.965 to 0.686) and in three of six categories (cheese, soft drinks, analgesics). It becomes the most accurate model in cheese (0.545 against 0.708 for LightGBM) and is level with the world model on Frat (0.686 against 0.681), but stays behind a model trained on the plan in the other categories. In bottled juice the plan made it slightly worse (0.940 to 0.990).

The trained plan-aware models also fall short in promoted weeks: the world model forecasts 48% to 88% of promoted-week sales and LightGBM 50% to 84%. Among the trained models, which one is more accurate depends on the category (Table 4): over all levels the world model is ahead in soft drinks (0.830 against 0.989) and analgesics (0.867 against 0.978), level in bottled juice, slightly behind in frozen entrees (0.485 against 0.478), and behind in cheese, snack crackers and on Frat; item by store LightGBM is level or ahead in every category. On Dominick's the world model's gains are at the totals, the opposite pattern to M5.

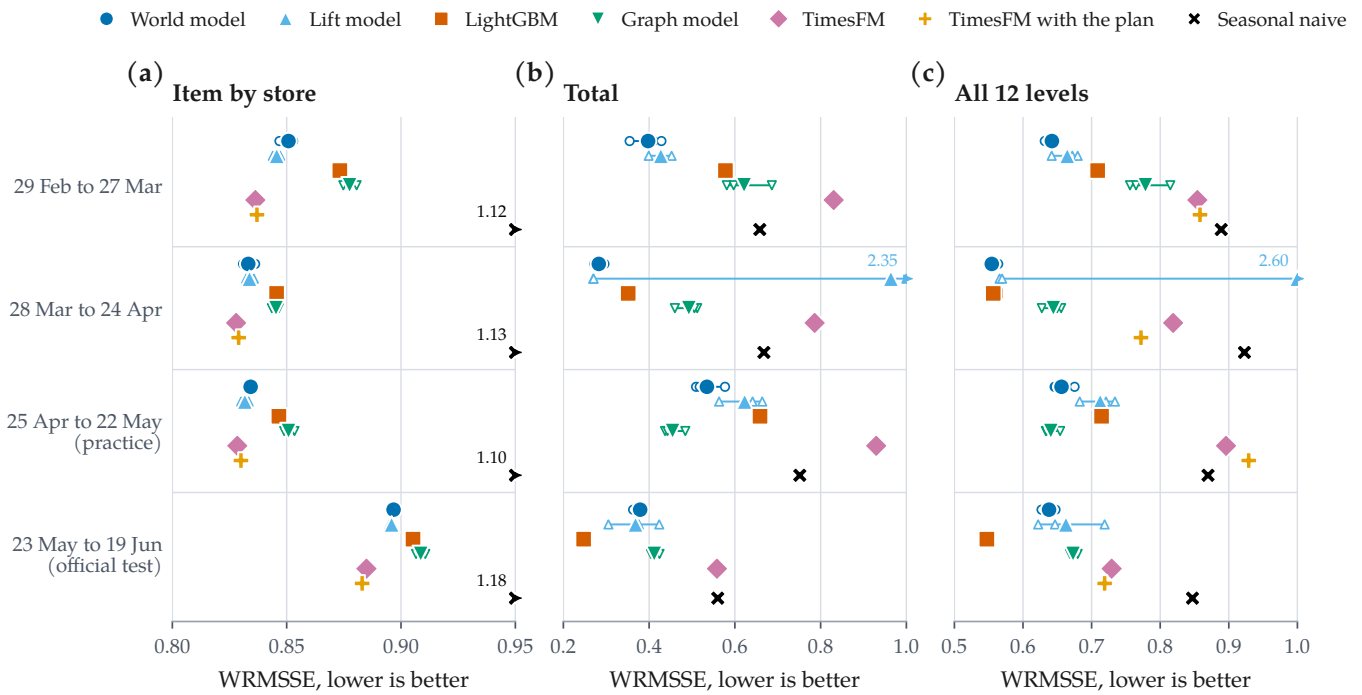


Figure 3: Item by store, TimesFM from history alone is the most accurate model in every month; at the total and over all levels, the plan-aware models are, and which one changes by month. WRMSSE on four M5 windows, lower is better. Large markers: mean over seeds; small open markers: each seed; bar: seed range. Values beyond the axis are drawn at its edge with their value. One lift-model seed failed in April (§5.5). **(a)** Item by store (30,490 series). **(b)** Chain total. **(c)** All 12 levels, the M5 score.

5.4 First-time price cuts

In the April M5 window, HOUSEHOLD_1_146 had sold almost nothing for a year at 5.98, then sold 120 units in four stores over a window that held a one-week cut to 4.50 (Figure 7b). Models working from history forecast almost nothing: LightGBM 3.1, 2.9, 3.3 units across seeds, TimesFM 7.8. The world model, which reads the cut through its plan, forecast 121, 600, 243; the graph model 59, 45, 55; the lift model 79, 206,886, 316.

Models that read the plan respond to a price change that history has not shown, and LightGBM, which sees price only against its own recent highest, responds little. How large the response should be is not fixed by the data: the world-model seeds differ by a factor of five, and one lift-model seed forecast over 200,000 units. For the same window, FOODS_3_443 in one store had a 68% cut and sold 4 units; every plan-reading model forecast more (the world model up to 26).

5.5 Negative and degenerate results

We report the failures at the same length as the results above. Two had a measured cause; the other changes and comparison models are in Table 5.

A lift-model seed that broke on five series. In April, lift-model seed 2 scored 2.603 over all levels against 0.566 and 0.570 for seeds 1 and 3, and forecast 1,413,724 units against 1,183,626 sold. Five item-store series hold all of the excess. Each had its first price cut in a year inside the window (the four HOUSEHOLD_1_146 stores above, and FOODS_3_443 in one store, 2.50 to 0.79 for three weeks). Seed 2 read the cuts as lifts up to the model's bound, about 22,000 times the item's level per day. Item by store, weighted by dollar sales, the failure is invisible (0.834); at the totals it dominates. The world model, with the same readout and the roll-forward, did not fail on these series in any seed.

Frat holiday weeks. On the Frat development windows the world model, lift model and LightGBM were level (0.406 to 0.421). On the final 8 weeks the world model fell to 0.681 against 0.549 for LightGBM, and every LightGBM seed beats every world-model seed. The final weeks are the holiday period, which the development windows did not contain (Figure 8). The world model's 8-week total is right (forecast against sales 1.000, LightGBM 0.937), but it is 15% to 17% under in the two December peak weeks and 24% over in the week after Christmas. LightGBM, with week-of-year and month inputs and training data that include the previous holiday season, follows that shape more closely.

A withdrawn claim. An earlier reading that the world model was more accurate than LightGBM on fast-selling



Figure 4: The order of the models reverses between the upper levels and item by store. WRMSSE at each of the 12 M5 levels, from the chain total (left) to item by store (right), one panel per window. Lines: mean over seeds; bands: seed range. Shaded columns: the two levels the decisions in §6 are made at.

items came from grouping pairs by their sales in the scored window, which pulls forecasts in high groups towards under and in low groups towards over for both models. Grouped by an earlier window (Figure 5b), the advantage is small and the same in every group. We do not make the claim.

Changes that did not become defaults. Table 5 lists every change screened and every comparison model that lost. Longer training, weekday and event inputs, a long-run level, a totals term in the loss, a count likelihood, a separate plan path in the state, feeding the model's own forecast into the roll-forward and item role labels either made no difference beyond seed noise or made the model worse. The graph model is level with LightGBM item by store, whose inputs it shares, so sharing information between related series added nothing on these inputs. TimesFM's default median output forecasts 27% to 31% below M5 totals, which makes its upper-level error worse than seasonal naive; its mean output, used above, does not have that problem.

6 Discussion

A retailer should choose a forecasting model by the decision it feeds, scored at that decision's level and over several periods (Table 6).

Two cautions follow from the failures. A direct lift readout can extrapolate a price cut it has never seen for an

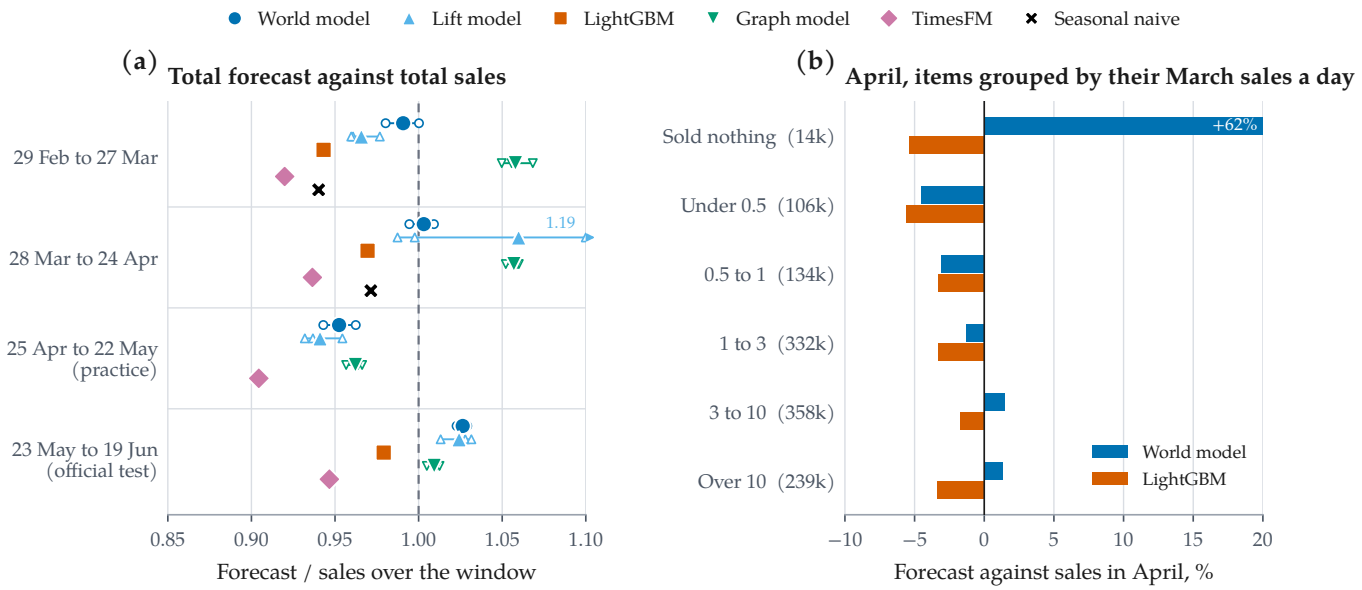


Figure 5: Whichever model gets the month's level closer wins the totals, and that changes by month. **(a)** Total forecast against total sales on each M5 window (1.0 is right); markers as in Figure 3. Seasonal naive's ratio was not recorded on the two later windows. **(b)** April, with each item-store pair grouped by its average daily sales in March (units sold in April in brackets), so that the grouping does not depend on the sales being scored: forecast against sales by group, world model and LightGBM, three seeds averaged.

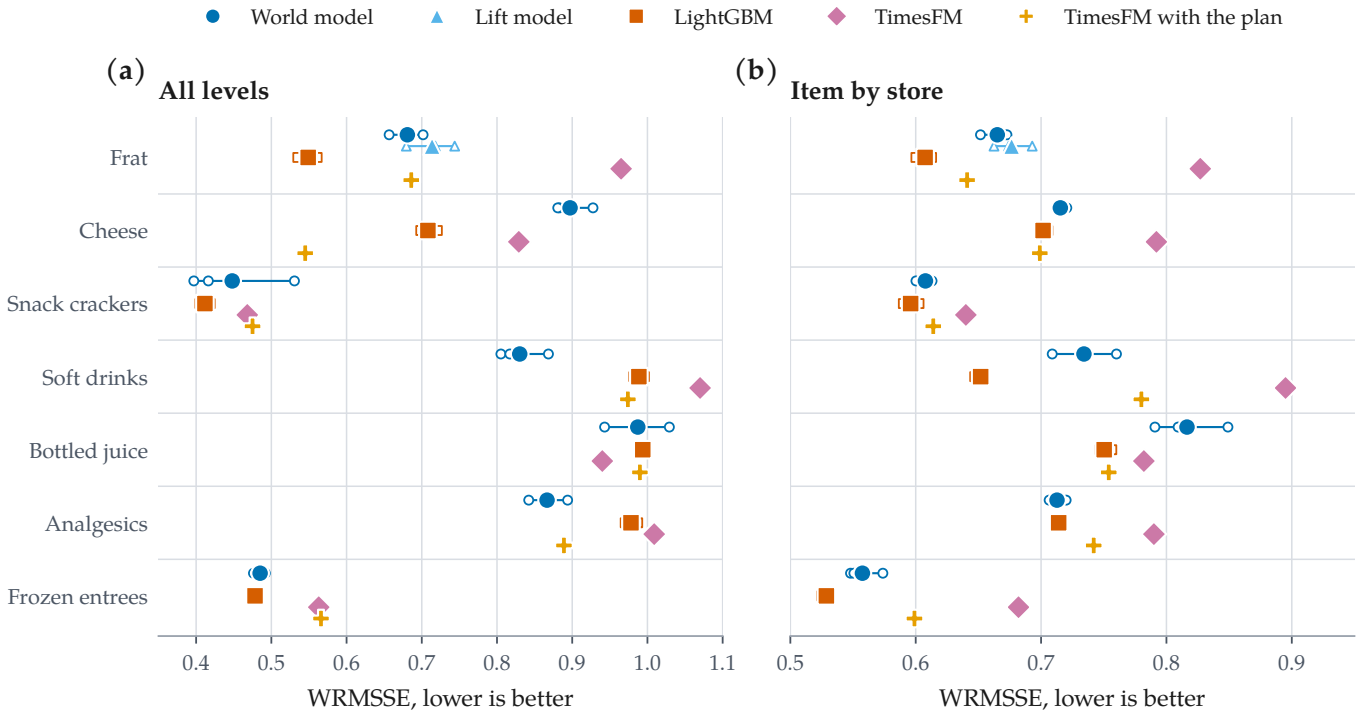


Figure 6: On the promotion-driven weekly data, TimesFM from history alone is behind both plan-aware trained models item by store in six of seven data sets; given the plan, it closes most of the gap. WRMSSE on the final 8 weeks, lower is better. World model, LightGBM and (Frat only) lift model: three seeds, markers as in Figure 3; TimesFM and TimesFM with the plan: one run. **(a)** All six levels. **(b)** Item by store.

item far beyond any sale, and whether it does depends on the seed; a plan-driven forecast needs a bound and a check on series whose plan falls outside their own history. And a forecast that is accurate item by store can be several percent off the category total, which is the number a merchant plans against.

The comparison also says what the world model's roll-forward adds and what it does not. Against the lift

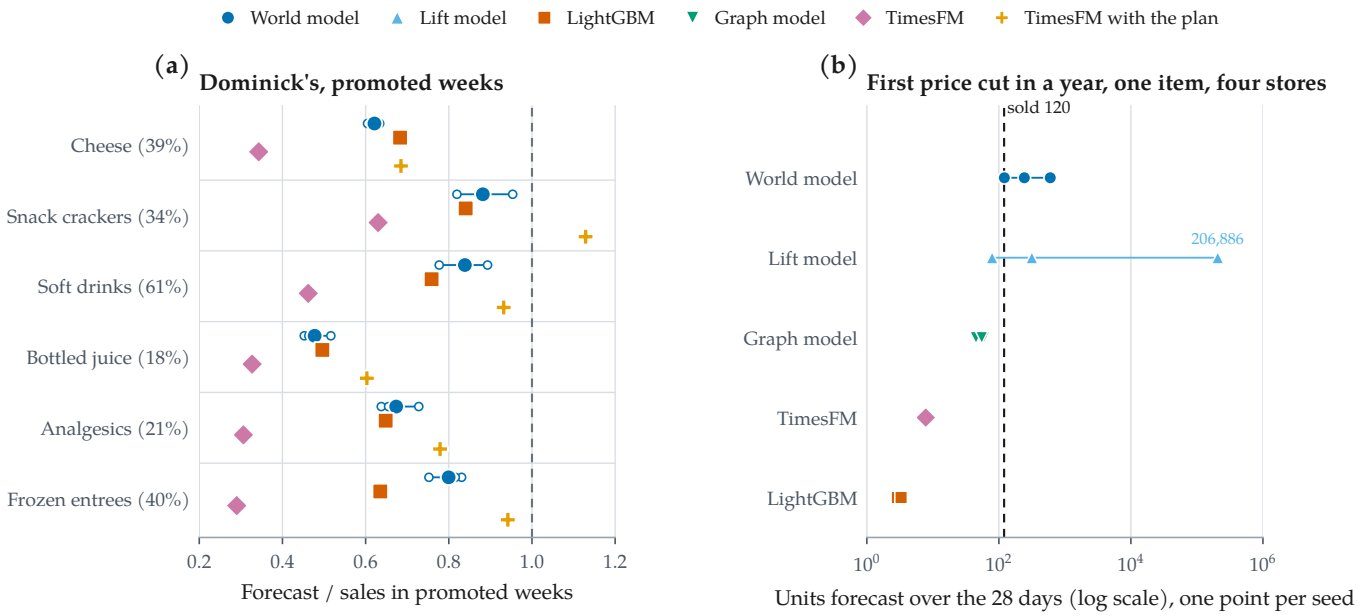


Figure 7: Without the plan a model cannot see a promotion or a price cut coming; with it, the direction is right and the size is uncertain. (**a**) Forecast against sales in promoted weeks of the final 8 weeks, by Dominick's category (promoted weeks' share of the category's units in brackets); 1.0 is right. (**b**) HOUSEHOLD_1_146 in the four M5 stores where it had its first price cut in a year (5.98 to 4.50 for one week) inside the April window: units forecast over the 28 days by each model, one point per seed, log scale; it sold 120.

Table 5: Every change and comparison model that did not become a default. Overall WRMSSE unless stated; "against" gives the default it was compared with on the same window. dev.: development windows before the final 8 weeks. Practice: the M5 window 25 April to 22 May 2016. Each row is a section of `results/README.md` with its run folders.

Change or model	Where tested	Result
Twice the training steps	Dominick's (6) and Frat, dev., 3 seeds	Better in 2, worse in 3, level or mixed in 2
Weekday as seven inputs	M5 practice, seed 1	0.653 against 0.646
Event kinds (with weekday)	M5 practice, seed 1	0.646 against 0.646
60,000 training steps	M5 practice, seed 1	0.642 against 0.646; gain below seed range 0.029
Item level against past-year level	M5 practice, seed 1	0.680 against 0.646
Totals term in the loss	Dominick's analgesics, dev.	0.819 against 0.683; total 9% under
Negative binomial likelihood	Frat, analgesics, dev.	Level
Plan split, latent check, near-week weighting	Frat, dev., 5 seeds	0.417 to 0.427 against 0.408
Own forecast fed into the roll-forward	Dominick's (6), dev.	Worse overall in 4 of 6
Item role labels	Dominick's (6), dev.	No effect on LightGBM; snack crackers 2.422 for the world model
Averaging three fits	M5 practice, seed 1	0.665 against 0.656 (kept on Dominick's)
Graph model over product attributes	M5, four windows, 3 seeds	Level with LightGBM item by store
TimesFM, median output	M5, four windows	Overall 1.71 to 1.88; totals 27% to 31% under
Lift model (no roll-forward)	M5, four windows, 3 seeds	One seed failed in April (2.603)

model, which differs only in the roll-forward, it is level or better over all levels in every M5 month and on Frat, and it did not fail on the first-time price cuts that broke a lift-model seed. Item by store it is not more accurate than the lift model, and on M5 neither is more accurate than a pretrained model that sees no plan at all.

7 Limitations and threats to validity

Windows and seeds. M5 has four final windows; Dominick's and Frat one each. On the M5 practice window LightGBM was run with one seed. Three seeds per trained model show the spread of training, not of the data.

Table 6: Each decision has its own level, and the results point to a different model for each.

Decision	Level	What the results say
Store orders and inventory for items whose price rarely moves	Item by store, day or week	A pretrained model from history alone is as accurate as or more accurate than plan-aware models (M5: most accurate in all four months)
Promotion, price change, supplier funding	Item by week, under a planned change	A model that reads the plan is needed; history alone forecasts 29% to 63% of promoted-week sales and does not respond to a first price cut
Category sales, units and margin plans	Category or chain, week or month	Judge on the level of the total; the world model was within 1% of total M5 sales in both confirmation months
Choosing a model	Any	Score at least three or four periods; one period reversed the ranking on M5

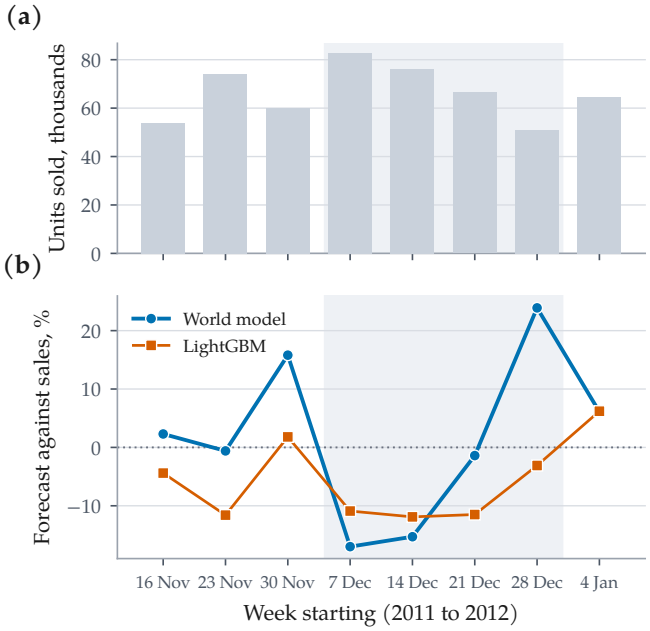


Figure 8: On Frat the world model gets the 8-week total right and misses the holiday weeks. (**a**) Units sold each week of the final window, all item-stores. (**b**) Forecast against sales each week, three seeds averaged. Shaded: the December weeks, absent from the development windows.

Comparison models added later. TimesFM, TimesFM with the plan and the graph model were added after the final tests, with settings fixed before their first run but without development windows of their own. Each TimesFM variant is one run, since it has no training randomness. TimesFM with the plan fits its linear plan model within groups of 2,048 series, a memory limit, so the fit is shared within each group.

The size of a plan response. The world model's response to a first-time price cut varied by a factor of five across seeds. The data do not identify it, and a deployed system would need a bound or a prior.

What the read-back shows. A high read-back R^2 shows the plan is in the world model's state, not that the forecast uses it correctly.

Public data only. The three data sets are public and were collected between 1989 and 2016. Promotion records differ between them (Dominick's records a promotion flag; Frat records display and feature separately), so the plan inputs are not the same across data sets.

8 Conclusion

Which forecasting model is most accurate depends on the level it is judged at, the month, and how much of the sales the plan moves. On M5 a pretrained model reading history alone is the most accurate item by store in every month, while plan-aware models are more accurate at the totals, and the official test month is the only one of four in which LightGBM wins overall. On promotion-driven data a model without the plan forecasts well under half of promoted-week sales in most categories. A world model that carries each item's state forward under the plan responds to price changes history has not shown, in the right direction and with an uncertain size. Retail forecasts should be chosen and reported by the decision they feed, at that decision's level, over several periods, with the losing models stated.

9 Reproducibility and compute

All code, configuration files and tests are in the repository. Every run records its code commit, configuration, data checksums and seeds in a manifest; the manifest and scores of the 110 runs read by this manuscript are committed under `results/promoted/`, and the manuscript's figures, tables and quoted numbers are generated from them by `paper/figs/make_figures.py` and `paper/figs/make_tables.py`. Numbers from runs not in that folder (TimesFM on Dominick's and Frat, TimesFM with the plan, and the analyses of saved forecasts) are copied into `paper/figs/rwdata.py` from `results/README.md`, which names each run folder. Forecast files and model checkpoints are kept outside the repository, with their checksums recorded in each run's scores. Runs used one NVIDIA Tesla T4 GPU on Google Colab: about 100 minutes per seed for the world

model on M5, 78 for LightGBM, 7.5 for the graph model and 4 for TimesFM.

10 Data and code availability

M5 is available from the competition's Kaggle page [1]; Dominick's from the Kilts Center for Marketing, University of Chicago Booth School of Business [6]; Breakfast at the Frat from dunnhumby [7]. Code: `https://github.com/Kar488/retail-world-models`.

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